

Solving for the Initial Phase (ϕ)

A Step-by-Step Method for $x(t) = X_m \sin(\omega t + \phi)$

When the initial conditions (x_0, v_0) do not correspond to peaks or equilibrium, students must use a rigorous algebraic approach.

1. The Starting Equations (at $t = 0$)

- Position:** $x_0 = X_m \sin(\phi)$
- Velocity:** $v_0 = \omega X_m \cos(\phi)$

2. The Ratio Method (Finding the Tangent)

To eliminate the unknowns X_m and ω and isolate ϕ , we divide the two equations:

$$\frac{x_0}{v_0/\omega} = \frac{X_m \sin(\phi)}{X_m \cos(\phi)} = \tan(\phi)$$

Therefore:

$$\tan(\phi) = \frac{\omega x_0}{v_0}$$

3. The Two Possible Angles

Since the tangent function has a period of π , the calculator's `arctan` function only gives one value (the "Principal Value"). There are always two possible angles:

- $\phi_1 = \arctan\left(\frac{\omega x_0}{v_0}\right)$
- $\phi_2 = \phi_1 + \pi$

4. Selecting the Correct ϕ (The Logic Test)

To pick the correct one, look back at your initial equations.

- Step A:** Look at $x_0 = X_m \sin(\phi)$. Since X_m is always positive, $\sin(\phi)$ must have the same sign as x_0 .
- Step B:** Look at $v_0 = \omega X_m \cos(\phi)$. Since ω and X_m are positive, $\cos(\phi)$ must have the same sign as v_0 .

Condition	$\sin(\phi)$ sign	$\cos(\phi)$ sign	Quadrant
$x_0 > 0, v_0 > 0$	+	+	Q1 ($0 \rightarrow \pi/2$)
$x_0 > 0, v_0 < 0$	+	−	Q2 ($\pi/2 \rightarrow \pi$)
$x_0 < 0, v_0 < 0$	−	−	Q3 ($\pi \rightarrow 3\pi/2$)
$x_0 < 0, v_0 > 0$	−	+	Q4 ($-\pi/2 \rightarrow 0$)

Example for Class

Given: $\omega = 10 \text{ rad/s}$, $x_0 = 3 \text{ cm}$, $v_0 = -20 \text{ cm/s}$, $X_m = 5 \text{ cm}$.

- $\tan(\phi) = \frac{10 \times 3}{-20} = -1.5$
- $\phi = \arctan(-1.5) \approx -0.98 \text{ rad (or } -56^\circ)$.
- Check signs:** x_0 is (+), v_0 is (−). This requires Quadrant 2.
- Our angle -0.98 is in Q4. We must add π to get into Q2.
- Final Answer:** $\phi = -0.98 + 3.14 = 2.16 \text{ rad}$.